# A projector, a trace and a norm have distinct jobs

WRITTEN PROOF / quadratic conjugation

$$\alpha = a + b\sqrt{2} \xrightarrow{\sigma} \sigma(\alpha) = a - b\sqrt{2}$$

$$E_{C_2}(\alpha) = \frac{\alpha + \sigma(\alpha)}{2} = a$$

$$\mathrm{Tr}_{L/\mathbb{Q}}(\alpha) = 2a$$

$$N_{L/\mathbb{Q}}(\alpha) = a^2 - 2b^2$$